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Sports injuries and game stakes: Concussions in the National Football League

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Abstract

The National Football League's regular-season games are not of equal importance: some games loom larger than others for determining a team's chance to qualify for the playoffs. We develop an incentive-based measure of the impact of winning a game on a team's qualification probability to study the relationship between stakes and injuries. We find teams are 24 percentage points more likely to suffer concussions in games where a win secures one team a playoff berth. This is the first evidence to support the risk-escalation hypothesis that injuries increase with a competition's stakes. We then discuss implications for sports injury prevention.

KEYWORDS

concussion, incentive, injury policy, injury prevention, National Football League, risk-escalation, sports injury

JEL CLASSIFICATION

Z28, I12, I18

1 | INTRODUCTION

Sports injuries are a major concern to athletes, medical experts, and policy makers. Scholars in sports medicine and sports management have made multiple proposals to minimize the injury cost of participating in sport, such as prevention, treatment, protective equipment mandate, or changes to a sport's rules (Finch, 2006). This existing approach tends to view injuries as unfortunate yet largely unavoidable accidents.

In this work, we investigate whether some sports injuries occur because athletes take risk in the pursuit of the pinnacle of success. According to the *risk-escalation hypothesis*, injuries should be more prevalent as competition stakes increase because athletes are willing to bear additional risk. Demonstrating that game stakes influence injury prevalence opens new horizons to frame injury policy in term of incentive design, under the general premise that it is possible to influence risk-taking choices (Cisyk & Courty, 2023a, 2023b).

We investigate whether injury costs depend on game stakes by using the incidence of concussions sustained during National Football League (NFL) regular-season games. Because concussions have a high potential of lifetime

Abbreviation: NFL, National Football League.

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consequences for their victims, they are of particular concern in most sports of the contact or collision varieties, as best exemplified by American football (DeHaven & Lintner, 1986).[1] Sports organizations have implemented policies to increase player safety (Deubert et al., 2017), legislators have passed laws to standardize concussion detection and diagnosis (Rotolo & Lengefeld, 2020), and players have initiated legal battles that threaten the financial viability of some professional sports (Bull, 2021).

Despite being the world's most valuable sports league with massive investments in injury detection, prevention, and treatment, the NFL still grapples with concussions. Moreover, the NFL offers rigorous reporting of injury events (at least relative to other sports) subject to strict monitoring and public-disclosure rules, and a mature literature studying confounding factors. Finally, the NFL offers an exemplary data laboratory: each of the league's 32 teams play 16 games during the regular season, some with high stakes, and others not.

We derive a gauge of game stakes that measures a team's incentive to win a game that we term the game's swing. The swing represents the increase in the probability that the team qualifies for the post-season playoffs from winning a given game. The swing value generalizes the notion of mathematical certainty (i.e., whether a team can still influence its qualification status) and expands on other measures of game importance used in the literature (Lahvička, 2015; Lei & Humphreys, 2013; Scarf & Shi, 2008; Schilling, 1994). Computing exact swing values requires considering all possible future path-dependent game outcomes, which is beyond computational possibility. Section 4 proposes an approximation of the swing value and demonstrates that this approximation performs well under two simple and testable conditions that are satisfied for a large fraction of observations in our sample.

Figure 1 reports the swing values, at each stage of the NFL regular season, as a function of past wins (vertical axis) and losses (horizontal axis). Although swing values are team- and game-specific, for the sake of the present argument, it helps looking at averages. As the season progresses, fewer games remain, and teams move further away from the

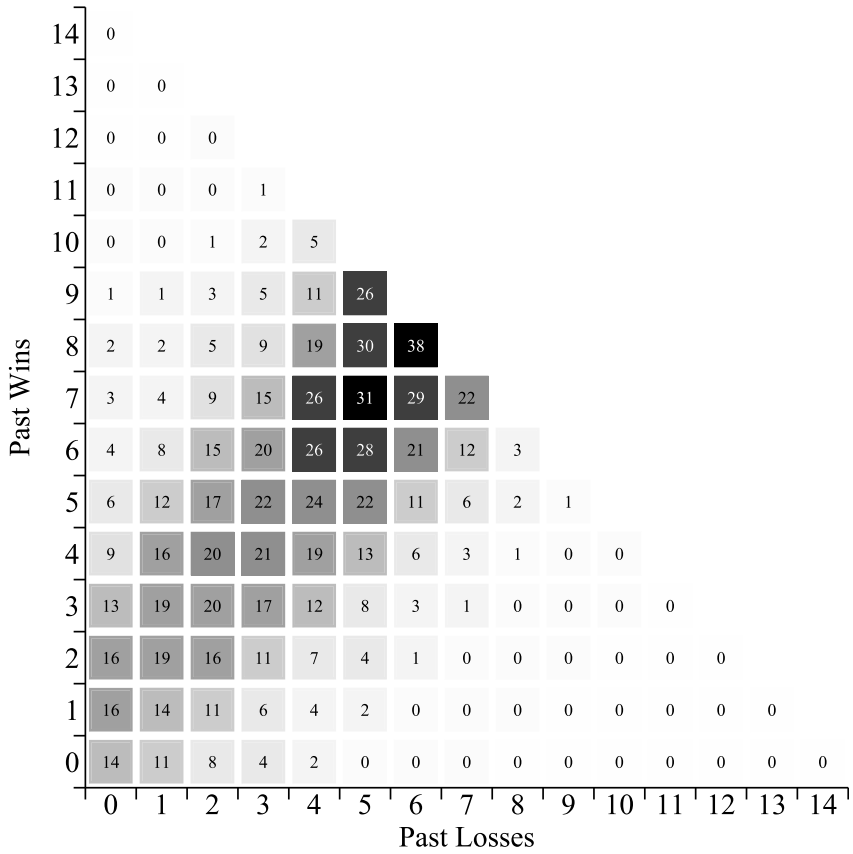

FIGURE 1 Average swing values as a function of past wins and losses from National Football League regular-season games, 2012–2015. *Note*: The swing value is computed for games 1–15 for each team and season using the definition of $\widehat{Z}_t^i$ in Equation (2). The values reported in the figure are team-season averages. Since the sum of wins and losses prior to game t is equal to $t-1$, past wins and losses can sum to a maximum of 14.

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lower-left cell (denoted as [0,0] indicating a team's first game). The figure reveals modest game stakes at the beginning of the season. Yet, as the season progresses, a team located slightly above the 45-degree diagonal (i.e., with a few more wins than losses) faces high incentives to win games. For example, consider a team that has already played nine games: if the team has won five and lost four, it faces a swing value of 24, saying that it increases its chance of qualifying for the playoffs by 24 percentage points by collecting its sixth win in its 10th game. In contrast, the swing values are at most 3 percentage points if the team has already won at least eight games or lost at least six: the playoff outcomes are nearly forgone conclusions regardless of future performance.

We find that the swing variable has a statistically significant impact on the incidence of concussions. While the average NFL team faces a 25% chance of observing at least one concussion in a game, this probability increases by 10 percentage points when a win increases one of the teams' odds of playoff contention by 40 percentage points.

We conduct a series of robustness tests. First, we eliminate observations with a large upper bound in the approximation error on the swing value. Second, we use the notion of mathematical certainty to improve the swing values. The third robustness approach leverages the sequential feature of the NFL playoff seeding process.[2] Finally, we consider different definitions of swing that account for a team's and its opponent's incentives to take risk.

Consistent with Hannah et al. (2019), we do not find any effect of game stakes on the incidence of concussions using mathematical certainly alone, which is a coarse measure of game stakes. In contrast, the swing variable captures finer, team-specific variation in incentives. We also find results consistent with the literature for confounding variables describing game conditions such as playing surface, team strategy, and weather. These findings contribute to the literature that studies the determinants of concussion in the NFL (e.g., see Heintz et al., 2020 for a recent work). Most importantly, our results have policy implications that we review in the conclusion.

The sports literature has long recognized that athletes respond to game rules, performance incentives, and to the general regulations that govern how to achieve superior performance (Frick, 2003). We argue in Section 2 that the behavioral responses described in the literature do not directly address the risk-escalation hypothesis, which applies to any sports where risk-taking increases both the chance from winning and the likelihood of injury. Section 3 introduces our case study of concussion events in the NFL and presents the data. Section 4 explains how we approximate the swing value. Section 5 presents the results and numerous robustness tests. The paper concludes with some implications of our work as well as some limitations.

2 | LITERATURE REVIEW AND HYPOTHESIS

A large literature has studied numerous determinants of injury in a variety of sports including American football. Part of the literature has looked at factors that are outside the control of athletes. For example, Haider et al. (2018) studies the role of weather while Fair and Champa (2019) looks at game rules. Injuries also happen when an athlete takes excessive risk, by for example, putting strain on the body, selecting strategies that involve body-contact or falls, or neglecting prevention. There is much evidence that such type of risk increases the chances of exceptional performance at the cost of higher chances of injuries (e.g., Bolling et al., 2020; Chen et al., 2019; Turner et al., 2004; Verhagen et al., 2010). Courty and Cisyk (2023a) speculates that athletes can increase their chance of success by taking more risk but suffer higher expected costs of injury. Athletes will assume more risk when the rewards from doing are higher. Our main hypothesis is derived from Lemma 1 in Cisyk and Courty (2023a)[3]:

Risk-escalation hypothesis: Injuries increase in likelihood during games with higher stakes.

The risk-escalation hypothesis is novel in that it directly links injury to game stakes.[4] There is much evidence that athletes respond in a variety of ways—including unexpected ones—to changes in game rules, that regulate contact and dangerous play, and to the adoption of safety measures such as mandatory protection equipment (Bradley, 2013; Potter, 2011). Such evidence shows that some injuries are caused by athletes' behavioral responses (Arias et al., 2011; Lybbert et al., 2012; Mack et al., 2021). There is also much evidence that athletes change effort and strategies in response to game stakes (Kahn, 2000). We are not aware, however, of any work that directly connects game stakes and injuries under the premise that athletes increase their risk of injury in response to a game's stakes.

To implement the risk-escalation hypothesis, one should focus on a type of injury that is caused by risk taking (among other factors), happens frequently, has significant health consequences, and for which the researcher has access to exogenous variation in the incremental gain from taking risk. We choose to focus on concussion because this type of injury has been widely studied and quality data about in-game injury is available as explained in the next section.[5] A common concern with injury data is under-reporting. Kroshus et al. (2015) suggest that only one in seven concussions

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sustained in amateur American football was diagnosed in 2012 using survey information from the National Collegiate Athletic Association. Others have argued that athletes in amateur sports may withhold from self-reporting their injuries for fear that doing so would prevent them from matriculating to the professional level (Brophy et al., 2009).[6] Under-reporting could cause the researcher to under-estimate the impact of stakes on injury risks and concussions.

Our work also contributes to the literature that uses the same concussion data used here to study the determinants of concussions in the NFL (e.g., Dai et al., 2018; Haider et al., 2018; Heintz et al., 2020; Myer et al., 2014; Teramoto et al., 2015, 2017).[7] Smoliga and Zavorsky (2017) argues that a limitation with this literature is confounding interpretation in the presence of omitted variables and limited exogenous variation. We follow this advice and use a panel with three categories of fixed effects and many control variables to reduce the risk of misinterpretation. Most importantly, we construct the swing variable: a theory-based estimate of the returns from winning a game. Closely related to our work, Hannah et al. (2019) uses the notion of mathematical certainty to split observations into three groups (teams that have already qualified for the playoffs, teams that are already eliminated, and the remaining teams that are still "in-the-hunt") and do not find any significant difference in concussion rates across groups. This approach is subject to the critique of Smoliga and Zavorsky (2017) to the extent that much of the variation in the three groups is explained by team heterogeneity. In contrast, we employ team fixed effects to control for unobserved team-heterogeneity, and leverage within-team changes in swing. The difference in our findings suggests that identification comes from variations within games that are mathematically uncertain (the observations around the diagonal in Figure 1).

3 | CASE STUDY AND DATA

3.1 | Case study

During our sample period, which covers the 2012 to 2015 seasons, each of the 32 NFL teams play 16 games during the regular season, typically between September and December of each year. At the end of the regular season, 12 of the 32 teams qualify to advance to the single-elimination playoff tournament to determine the season's champions in the finale known as the Super Bowl. Playoff qualification is largely determined on a team's win-loss record at the end of the season with other considerations accounted for when ties occur (see Supporting Information S1: Appendix 2 for details).

Many regard the playoff games as the pinnacle of American football with Super Bowl viewership regularly reaching over 110 million households from 2012 to 2015. Players are remunerated for participating in playoff games. For the final season of our sample (i.e., 2015) players are compensated a minimum of $23,000 and a maximum of $198,000 for playing in the playoffs (amounts vary by the extent of the team's advancement through the single-elimination tournament). For a player earning the league's minimum salary of $435,000, these bonuses range from five to 45%.[8]

Concussions are a common injury in American football with devastating life-changing consequences (Heintz et al., 2020; Navarro et al., 2017). After denying the importance of concussions for years, the NFL has faced multiple lawsuits from injured players (Bull, 2021). To increase player safety, the NFL has begun, staring in the early 2000's, to implement a set of guidelines covering concussion spotters, on-field medical professionals, film reviews, examination, treatment, and return-to-play rules. Supporting Information SI: Appendix 1 provides a list of rule changes implemented during our sample period designed to limit injuries, including concussions. The presence of such rule changes requires the estimation strategy to consider temporal effects, as we discuss in the Identification section below.

3.2 | Data

We combine data from numerous publicly available sources. Our variable of interest is the concussion events sustained in regular-season NFL games, as detailed on official NFL injury reports and collected and summarized by PBS Frontline Concussion Watch.[9] The data contain information on the individual, position, team, the date of the concussion, the number of games the player was inactive (if any), and a brief description of the event. The information is summarized from official weekly injury reports released by each NFL team. These reports list all players that have reportable injuries, the injured body part, the player's level of participation in the weekly practices, and their expected status for the next game (see Data Supporting Information S1: Appendix 1 for details). Only injuries containing the terms "concussion" or "head" are selected.[10]

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We collect game-specific information on all games for four seasons (2012–2015), yet exclude the last game (i.e., game 16) of each team in each season, as teams that do not qualify for the post-season playoffs are not required to submit reports for injuries suffered during their final game. We also exclude 12 instances (24 team-games) where the game is played at a "neutral-site" location (i.e., London, Mexico City, and Toronto) as each match has idiosyncratic features that may influence injury risk (e.g., playing surface, travel requirements, days of rest) thus preventing statistical inference (Smoliga & Zavorsky, 2017). The number of concussions per season during our sample period ranges from 123 to 199, for a total of 645 concussions during a span of 1024 games. After our exclusions, we are left with 563 concussions in 948 games. At least one concussion occurred in 43% of games in our sample (multiple concussions occur in roughly one quarter of the games).

We supplement this information with game characteristics including the temperature, precipitation, the material and altitude of the playing surface, and the offensive and defensive strategies used by each team, collected from publicly available, online sources. We also calculate the distance the visiting team traveled and the number of days of rest since each team last participated in a game. Table 1 reports each variable, and its definition, source, and descriptive statistics. Finally, we collect data from the number of wins of each team and its playoff qualification outcome from the 2002 to 2019 seasons. Section 4.1 uses this information to estimate the probability to qualify to playoff conditional on season's wins.[11]

At the end of each season, the top 12 teams, ranked by win-loss record, qualify to participate in the playoffs.[12] The next section presents a general and computationally manageable framework to approximate the swing values for the 256 games that take place in each season. The framework could be applied in other sports' competition where contestants (teams, in our application) compete for a reward (playoff admission) that is distributed using a single outcome measure (final win count).

4 | SWING VALUES AND IDENTIFICATION

The swing value measures a team's game stakes as the marginal impact of winning a game on the team's chance to earn a playoff berth.

Definition. The swing value of team i for game t is the expected increase in its chance of earning a playoff berth due to winning game t.

This incentive-based definition differs from past works saying that a game is important if its outcome has a large impact on end-of-season-outcomes (Geenens, 2014; Schilling, 1994). For example, Lahvička (2015) defines importance as the "strength of relationship between the match result and a given season outcome." According to Lahvička, a game that features a strong team against a weak one is unimportant because the outcome is almost certain. In contrast, we assert that such a game can still have a large swing value for one (or both) of the teams, as will be the case if the outcome of the game has a large impact on a team's chance to qualify for the playoffs.

To formalize our definition, say that team i has already won $w^i_{1,t-1}$ of the $t-1$ games it has played up to game t, and expects to win n of the $T-t$ games that remain after game t with probability $\Pr\left(\tilde{w}^i_{t+1.T}=n\right)$, where $\tilde{w}^i_{t+1,T}$ is the random variable that is the cumulative number of games team i expects to win after game t. In our notation, the absence of a tilde on $w^i_{\cdot}$ indicates realized game outcomes, while the presence of one indicates that future game outcomes are not yet known. Let X^i_{t-1} denote team i's set of attainable final win counts in the event team i loses game t. It ranges from $w^i_{1,t-1}$ (the team's current win count) to $w^i_{1,t-1}+T-t$ (the team's final win count if it were to win all T–t remaining games). Team i's swing value for game t is:

$$Z^i_t=\sum\nolimits_{W\in X^i_{t-1}}\Pr\left(\tilde{w}^i_{t+1,T}=W-w^i_{1,t-1}\right)\left(A^i_{t-1}(W+1)-A^i_{t-1}(W)\right) \tag{1}$$

where $A^i_{t-1}(W)$ is team i's expected chance to earn a playoff berth if it ends the season with W wins, where the expectation is computed using all information available just before game t is played. To explain, ignoring the outcome of game t, team i expects to end the season with one of the $T-t+1$ final win counts from the attainable set X^i_{t-1}. The

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TABLE 1 Descriptive statistics of NFL regular-season games, 2012–2015.

Variable	Definition	Source	Mean	St. Dev.	Min	Max
Concussion	Takes a value of one in observations where at least one player of the observed team sustains an in-game concussion, zero otherwise	NFL injury reports (PBS Frontline Concussion Watch)	0.253	0.435	0.000	1.000
Swing	Difference in playoff probability if team wins less playoff probability if team loses	Elo values from FiveThirtyEight	0.166	0.085	0.000	0.414
Distance traveled	Distance between away and home stadia in miles (takes a value of one if observed team is home team)	Google Maps	474.041	655.916	1.000	2722.410
Turf	Takes a value of one if game is played on artificial surface, zero otherwise	Pro-Football-Reference.com	0.440	0.497	0.000	1.000
Altitude	Stadium's height above sea level in feet	Worldwide Elevation Map Finder	519.201	898.275	1.000	5197.000
Days of rest	Number of days since observed team last played	Pro-Football-Reference.com	7.659	2.214	4.000	15.000
Rival	Takes a value of one if observed team's opponent is within the same conference and division, zero otherwise	Pro-Football-Reference.com	0.335	0.472	0.000	1.000
Nationally televised	Takes a value of one if game is nationally televised, zero otherwise	Sports Media Watch	0.214	0.410	0.000	1.000
Air Coryell	Team uses the "Air Coryell" offense scheme (week invariant; team-season variant only)	Pro-Football-Reference.com	0.257	0.437	0.000	1.000
Erhardt-Perkins	Team uses the "Erhardt-Perkins Offense" offense scheme (week invariant; team-season variant only)	Pro-Football-Reference.com	0.204	0.403	0.000	1.000
West Coast	Team uses the "West Coast Offense" offense scheme (week invariant; team-season variant only)	Pro-Football-Reference.com	0.430	0.495	0.000	1.000
Other offense	Team uses some other offense scheme (week invariant; team-season variant only)	Pro-Football-Reference.com	0.110	0.313	0.000	1.000
3–4 defense	Team uses the "3–4" defense alignment (week invariant; team-season variant only)	Pro-Football-Reference.com	0.470	0.499	0.000	1.000
4–3 defense	Team uses the "4–3" defense alignment (week invariant; team-season variant only)	Pro-Football-Reference.com	0.530	0.499	0.000	1.000
Air Coryell, opponent	Opponent uses the "Air Coryell" offense scheme (week invariant; team-season variant only)	Pro-Football-Reference.com	0.257	0.437	0.000	1.000
Erhardt-Perkins, opponent	Opponent uses the "Erhardt-Perkins Offense" offense scheme (week invariant; team-season variant only)	Pro-Football-Reference.com	0.204	0.403	0.000	1.000
Air Coryell, opponent	Opponent uses the "West Coast Offense" offense scheme (week invariant; team-season variant only)	Pro-Football-Reference.com	0.430	0.495	0.000	1.000
Other offense, opponent	Opponent uses some other offense scheme (week invariant; team-season variant only)	Pro-Football-Reference.com	0.110	0.313	0.000	1.000
3–4 defense, opponent	Team uses the "3–4" defense alignment (week invariant; team-season variant only)	Pro-Football-Reference.com	0.470	0.499	0.000	1.000

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TABLE 1 (Continued)

Variable	Definition	Source	Mean	St. Dev.	Min	Max
4–3 defense, opponent	Team uses the "4–3" defense alignment (week invariant; team-season variant only)	Pro-Football-Reference.com	0.530	0.499	0.000	1.000
Temperature	Air temperature in Fahrenheit	MeteoStat, Weather Underground	60.172	15.314	8.060	93.920
Closed roof	Game is played under a roof/domed stadium	MeteoStat, Weather Underground	0.253	0.435	0.000	1.000
No precipitation	Game is played without any precipitation	MeteoStat, Weather Underground	0.621	0.485	0.000	1.000
Light rain	Game is played during light rain	MeteoStat, Weather Underground	0.078	0.268	0.000	1.000
Moderate rain	Game is played during moderate rain	MeteoStat, Weather Underground	0.025	0.157	0.000	1.000
Heavy rain	Game is played during heavy rain	MeteoStat, Weather Underground	0.007	0.086	0.000	1.000
Light snow	Game is played during light snow	MeteoStat, Weather Underground	0.011	0.102	0.000	1.000
Moderate snow	Game is played during moderate snow	MeteoStat, Weather Underground	0.004	0.065	0.000	1.000

Note: Sample excludes all "neutral-site" games, and games played during final week of each season. Number of observations = 1896.

summation in Equation (1) considers these $T - t + 1$ possibilities separately. When team i wins no game after game t, for example, the marginal benefit of winning game t is $A^i_{t-1}(w^i_{1,t-1} + 1) - A^i_{t-1}(w^i_{1,t-1})$.

Computing the probability $\Pr(\tilde{w}^i_{t+1,T} = n)$ is manageable because it requires forecasting $T - t$ game outcomes (i.e., the games played by team i after game t) corresponding to 2^{T-t} possible state-dependent forecasts, in the case where a team's future win probabilities depend on the team's previous wins. Difficulty remains in the computation of the expected chance to earn a playoff berth $A^i_{t-1}(W)$. The allocation of playoff seeds at the end of the season depends on the final win counts of all teams: team i must consider how each team in the league will perform in all of their own $T - t$ remaining games. With 32 teams in the league, this is not computationally manageable, at least at the beginning of the season, when t is small.

To address this computational challenge, we propose to replace the team-interim payoff functions, $A^i_{t-1}(W)$, with the unconditional payoff function, denoted $A(W)$, and computed as the probability that a team ending the season with W wins receives a berth. Stated differently, $A(W)$ is the average over all teams of $A^i_0(W)$. It can be computed if one observes the playoff berths for multiple seasons, which is the case in our application because the playoff qualification rules has remained unchanged for 18 consecutive seasons (from 2002 to 2019). For example, we observe 60 instances during this period where a team ends its season with nine wins and in 23 of these instances the team earns a playoff berth. We obtain $A(9) = \frac{23}{60} = 0.38$. We approximate the swing value as follows:

$$\hat{Z}^i_t = \sum\nolimits_{W \in X^i_{t-1}} \Pr\left(\tilde{w}^i_{t+1,T} = W - w^i_{1,t-1}\right)\left(A(W+1) - A(W+x)\right) \tag{2}$$

To illustrate, take the extreme case when the incentive zone is a singleton, $Z = \{W^*\}$. When this is the case, the approximation of the swing value is exact. To see why, note that the team-interim payoff function and the unconditional payoff function are equal to a single-stepped function, which is equal to zero up to W^* and one thereafter. The swing value is equal to the probability that the team hits the threshold value W^*, $\hat{Z}^i_t = Z^i_t = \Pr(\tilde{w}^i_{t+1.T} = W^* - w^i_{1,t-1})$. Beyond this trivial case, the approximation of the swing value is perfect when two simple conditions hold. Let $Z = \{\underline{z}, .., \overline{z}\}$ denote the incentive zone: the set of final win counts such that one additional win increases the probability of making the playoffs: $A(W+1) > A(W)$ for $W \in Z$. The first condition says that the incentive zone is a subset of the attainable zone.

Condition 1: $Z \subset X^i_{t-1}$.

The second condition says that based on all information available prior to game t, team i is equally likely to achieve any attainable final win count within the incentive zone.

Condition 2: $\Pr(\tilde{w}^i_{t+1.T} = W - w^i_{1,t-1}) = k^i_t$ for any $W \in Z \cap X^i_{t-1}$.

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Condition 2 trivially holds when the incentive zone is a singleton. Note that both conditions are team-game specific. Each condition may hold for some team-game and not for others. We now establish our main result:

Proposition 1 *The approximated swing, $\hat{Z}_t^i$, is identical to the true swing, Z_t^i for any (i,t) such that conditions 1 and 2 hold.*

The proposition is proved in Supporting Information S1: Appendix A3 which also offers additional discussion. Because conditions 1 and 2 are team-game specific, we can ignore the observations that do not satisfy these conditions.

4.1 | Payoff function, conditions 1 and 2, and swing values

The shape of the payoff function is important for several reasons. First, a payoff function that closely resembles a step function, even though not exactly, will likely deliver good approximation of the swing values (because conditions 1 and 2 are more likely to hold). Second, the shape of the payoff function also influences whether the swing value varies. Specifically, the swing values are constant when the payoff function is a linear function of W: teams face the same incentives throughout the season, and it is not possible to identify the effect of swing value on injuries.

Figure 2 plots the estimated values of $A(W)$ using data from the 2002 to 2019 NFL seasons.[13] The blue dots report the sample probabilities while the back curve plots the fitted values using a logistic regression. As expected, both plots show that the payoff function is weakly increasing in $W_{i,T}$. A team that has won seven or fewer games at the end of the season has effectively zero chance to qualify for the playoffs ($A(W) = 0$) while a team that has won 11 games or more is certain to do so ($A(W) = 1$). Since the payoff probability increases approximately between seven and 11 wins, the incentive zone is $Z = \{7,8,9,10\}$. The non-linearity in the payoff function implies that the reward from winning one additional game is not constant: a team has a strong incentive to win one more game if it otherwise expects to win 10 games by the end of the season, two more games if it expects to win nine, and so on and so forth.

Next, we investigate when conditions 1 and 2 hold. The function $\Pr(\tilde{w}_{t.T}^i = n)$ is never exactly constant over Z. Thus, we implement condition 2 using an approximation. We select only team-game observations for which the function $\Pr(\tilde{w}_{t.T}^i = n)$ varies by at most 0.2 over the incentive zone. The set value 0.2 implies that the measurement error, $|Z_{it} - \hat{Z}_{it}|$, will be at most 0.2.[14] Figure 3 plots the fraction of observations that do not satisfy either condition. Conditions 1 and 2 hold for all teams in each season up to the seventh game and at least half up to the tenth. Overall, these conditions hold for 67% of the observations.

According to Table 1, the swing value varies between zero and 0.41, with a mean value of 0.17 and a standard deviation of 0.9. Figure 4 depicts swing values for games 1–15, broken down into five categories akin to Figure 1 of Lahvička (2015). Swing values vary at each stage of the season and as the season progresses.

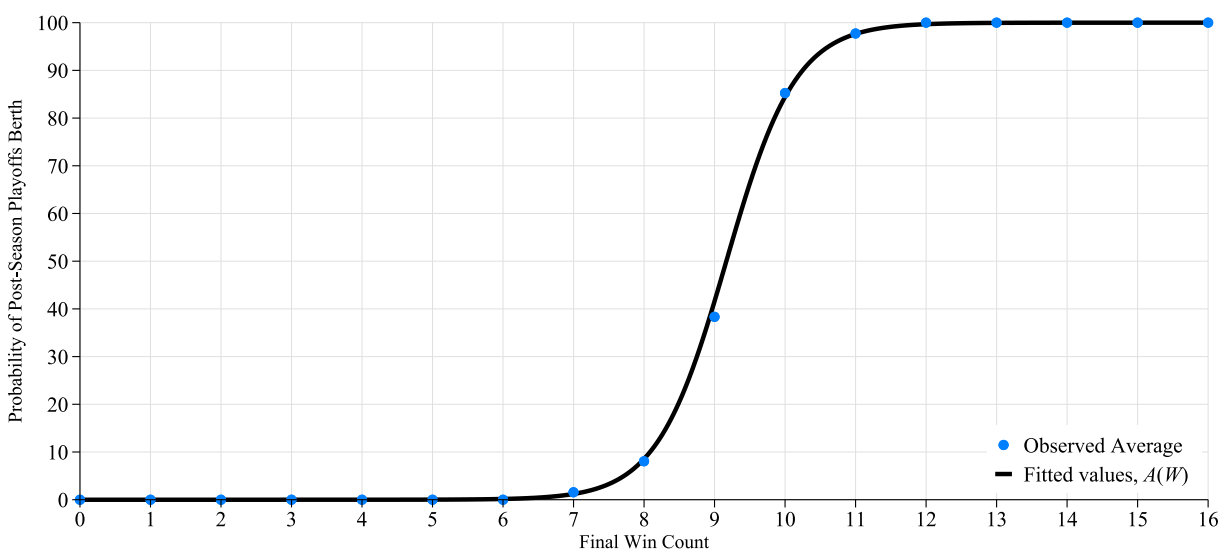

FIGURE 2 The payoff function (probability of earning a playoff berth as a function of total wins). Seasons 2002–2019.

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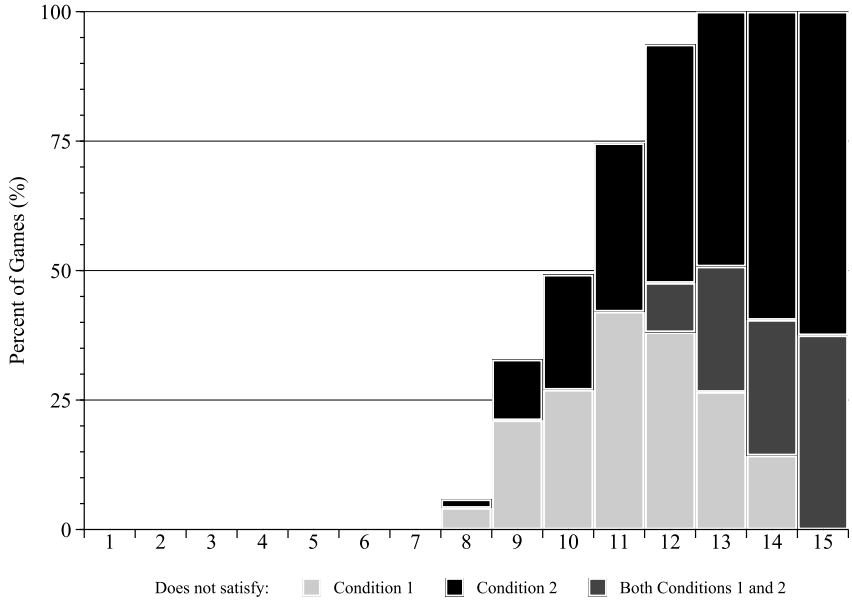

FIGURE 3 Violation of condition 1 and 2 by game.

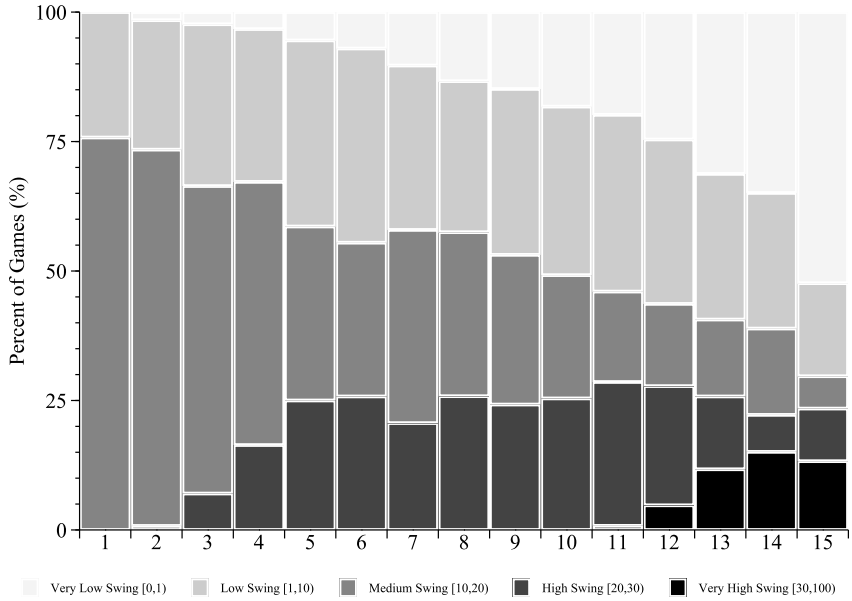

FIGURE 4 Variations in swing value within and across round.

Swing values vary because of luck and skill (Mauboussin, 2012): past win counts, $w^i_{1,t-1}$, are influenced by "luck" and future win counts, $\tilde{w}^i_{t+1.T}$, are random variables that are team-specific and liable to change over the season (teams' rosters change over the season as some players stand out and others disappoint). Prior to the first game of the season, the swing values vary by team only because of differences in team skills. A high-quality team that is likely to qualify for the playoffs faces little incentive to take risk. The same holds for a low-quality team that is unlikely to qualify. Marginal, or medium-quality, teams, however, can increase their chances to qualify for the playoffs by taking risk to get ahead of other medium-quality teams. After the first game, the swing values also differ because teams have different cumulated past wins. For example, two *identical* teams at the beginning of the season may face very different incentives in the middle of the season due to *luck* in their first few games ($w^i_{1,t-1}$ determines whether a team expect to end the season in the incentive zone).

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4.2 | Identification

Let C_{it} take a value of one if a player from team i sustains a concussion during game t, and zero otherwise, and let m_{it} denote team i's opponent in game t. We estimate the conditional probabilities

$$\Pr\left(C_{it} = 1 \middle| \hat{Z}_{it}, \hat{Z}_{m_{it}t}, X_{it}, S, T, G\right) \tag{3}$$

where X_{it} is a set of control variables described in Section 3.2, S are season fixed effects controlling for change over years in concussion occurrence and game rules, T are team fixed effects controlling for heterogeneity in team strength and strategies, and G are game fixed effects controlling for seasonal changes that influence injuries. $\hat{Z}_{it}$ is team i's swing for game t (defined in Equation 2) and $\hat{Z}_{m_{it}t}$ is the swing of team i's opponent.

According to the risk-escalation hypothesis, teams take more risk when facing higher swing values. Because American football is a contact sport, a team that takes more risk increases the chance that it suffers a concussion as well as the chance that its opponent does too. Thus, we expect a team's incidence of concussion to increase with its own swing ($\hat{Z}_{it}$) and with its opponent's swing ($\hat{Z}_{m_{it}t}$). We have explored numerous specifications with different approaches to include $\hat{Z}_{it}$ and $\hat{Z}_{m_{it}t}$. Although the coefficient estimates have the expected signs and reasonable magnitudes, they are not always significant, with the exception of the case where we use a single swing variable that is equal to the maximum of own and opponent swing, $\max\left(\hat{Z}_{it}, \hat{Z}_{m_{it}t}\right)$. The core of the result section uses this definition, and we discuss alternative variables in the robustness section.

Random variations in game outcomes generate variations in the incentives to take risk, or to face a team that is likely to take risk, and this establishes the foundation for our identification strategy. Still, one must consider other mechanisms that could generate an association between the swing variable and injury. Team strategy could influence both injury and the likelihood that a team stays within playoff contention late within a season. For example, consider a team that employs a strategy that has a high chance of injury in addition to a high variance of success (the team can do extremely well and win most games or extremely poorly and lose most). This generates an opposite prediction to ours. The teams that select this strategy have low swing values and high injury rates relative to the other teams (high swing values and low injury rates). Team fixed effects control for unobserved team characteristics, including team strategies.

Beyond this, it is not clear why injury outcomes should be associated with swing values. Clearly, injury outcomes lower future expected performance if they constrain the set of players that can be fielded, but the impact on swing is ambiguous.[15] Although we cannot control for unobserved changes in team strategies that could influence both injury rates and swing values, we investigate the issue empirically in Supporting Information S1: Appendix A5. We find no relation between past injury outcomes on future swing values. This suggests that there is no causal link from injuries toward swing.

5 | RESULTS

We first discuss the results of estimating Equation (3). Then, we check whether the predictions of the control variables are aligned with the literature, and most importantly, we compare the magnitude of the swing coefficient with the magnitude of the control variables that have received attention in earlier works. Finally, we demonstrate that the main conclusion survive a barrage of robustness checks.

5.1 | Effect of swing on concussion

Tables 2 and 3 display the results of estimating Equation (3) using a linear probability model. The main conclusions do not change if we instead use a Probit model. Table 2 presents multiple specifications with different sets of fixed effects to address the problem of confounding explanation documented in the literature (Smoliga & Zavorsky, 2017). Table 3 adds the standard control variables used in the literature. Our main hypothesis relating to risk escalation says that the coefficient estimate on the swing variable should be positive and significant.

We start with Table 2. The top panel reports the results with observations that satisfy Conditions 1 and 2 (restricted to 67% of the observations) while the bottom panel uses the entire sample. Each panel reports five specifications with no fixed effects, then, unilaterally, team, season, and game fixed effects, and finally all three fixed effects together. We start

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TABLE 2 Impact of swing on concussions with fixed effects only.

Dependent variable: Concussion indicator	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**
Swing	0.175	0.124	0.216*	0.263**	0.256*
	(0.121)	(0.132)	(0.127)	(0.123)	(0.131)
Game FE	No	Yes	No	No	Yes
Season FE	No	No	Yes	No	Yes
Team FE	No	No	No	Yes	Yes
Observations	1282	1282	1282	1282	1282
Adjusted R^2	0.000	0.008	0.006	0.003	0.018
Dependent variable: Concussion indicator	**(6)**	**(7)**	**(8)**	**(9)**	**(10)**
Swing	0.140	0.137	0.184*	0.208*	0.254**
	(0.111)	(0.115)	(0.108)	(0.108)	(0.107)
Game FE	No	Yes	No	No	Yes
Season FE	No	No	Yes	No	Yes
Team FE	No	No	No	Yes	Yes
Observations	1896	1896	1896	1896	1896
Adjusted R^2	0.000	0.007	0.005	0.007	0.019

Note: *$p < .1$; **$p < .05$; ***$p < .01$. Standard errors clustered by team in parentheses. Sample excludes all "neutral-site" games, and games played during final week of each season; sample for models (1) through (5) also excludes games that do not satisfy both Conditions 1 and 2.

with the restricted sample. Column 1, without any fixed effects, says that the swing variable is positive (coefficient of 0.18) but not statistically significant. The same holds when we add game fixed effects in column 2. Adding season or team fixed effects increases the coefficient estimate (to 0.22 and 0.26 respectively) and increases its statistical significance to a marginal level (10%) in the former case and a conventional level (5%) in the latter case. As previously explained, team fixed effects control for the heterogeneity in teams' skill levels and strategies. It is thus not surprising that the inclusion of the team fixed effects increases the significance of the swing variable. The impact of season fixed effects suggests that there may be some heterogeneity in concussion rates across the seasons of our sample due to, for example, rule changes that alter the costs or benefits of risk taking as discussed in Section 3.1. Column 5 includes all three fixed effects. The coefficient estimate largely remains the same as with team fixed effects alone. Overall, the swing variable is consistently associated with a positive coefficient implying that concussions are positively correlated with game stakes as predicated under the risk-escalation hypothesis.

We repeat the analysis with the full sample and observe similar results. In most specifications, the coefficient on swing is lower by a few percentage points in the full sample relative to the restricted one. This holds for the specifications with fixed effects in Table 2 and the specifications that add the controls in Table 3. One explanation is that the full sample adds observations that have measurement error.[16] In the specification with all three sets of fixed effects, however, the coefficient estimate is almost the same in the two samples.

Once adding in the confounding control variables, as is done in Table 3, the estimated coefficient of swing remains positive, significant, and with similar magnitude. The coefficient estimate indicates that a game that guarantees one of the two teams a playoff qualification with a win, but exclusion with a loss (i.e., a swing value of one), significantly increases the risk of having at least one concussion by 24 percentage points.

5.2 | Control variables

Table 3 presents four specifications with control variables. All specifications include team and season fixed effects. The four columns use different samples (full or restricted) and include or exclude game fixed effects. We cover here only a few key findings and argue that the impact of the swing variable on concussions is large compared to the impact of control variables that have received much attention in the literature.

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TABLE 3 Impact of swing on concussions with control variables.

Dependent Variable: Concussion indicator	(1)	(2)	(3)	(4)
Swing	0.307**	0.248*	0.242**	0.237**
	(0.134)	(0.138)	(0.109)	(0.108)
ln(Distance traveled)	0.006*	0.006*	0.007***	0.007***
	(0.003)	(0.003)	(0.002)	(0.002)
Turf	0.064**	0.078***	0.062***	0.075***
	(0.028)	(0.028)	(0.023)	(0.021)
ln(Altitude)	0.003	0.005	−0.001	0.002
	(0.006)	(0.006)	(0.005)	(0.005)
ln(Days of rest)	0.065	0.049	0.023	0.015
	(0.044)	(0.050)	(0.033)	(0.036)
Rival	0.015	0.011	−0.006	−0.013
	(0.023)	(0.022)	(0.023)	(0.022)
Nationally televised	−0.007	−0.005	−0.022	−0.019
	(0.031)	(0.032)	(0.030)	(0.030)
Other offense (base)				
Air Coryell	−0.058	−0.058	−0.048	−0.046
	(0.067)	(0.067)	(0.038)	(0.038)
Erhardt-Perkins	0.088	0.090	0.057	0.056
	(0.062)	(0.062)	(0.041)	(0.041)
West Coast	0.022	0.021	−0.005	−0.005
	(0.047)	(0.046)	(0.032)	(0.032)
3–4 defense (base)				
4–3 defense	0.012	0.017	0.032	0.033
	(0.043)	(0.043)	(0.035)	(0.034)
Other offense, opponent (base)				
Air Coryell, opponent	−0.080**	−0.080**	−0.086***	−0.086***
	(0.039)	(0.039)	(0.028)	(0.028)
Erhardt-Perkins, opponent	−0.041	−0.042	−0.042	−0.042
	(0.053)	(0.051)	(0.039)	(0.039)
Air Coryell, opponent	−0.039	−0.039	−0.046	−0.044
	(0.049)	(0.047)	(0.035)	(0.034)
3–4 defense, opponent (base)				
4–3 defense, opponent	0.010	0.010	0.019	0.017
	(0.021)	(0.021)	(0.022)	(0.021)
Temperature	−0.003**	−0.002	−0.003***	−0.002
	(0.001)	(0.002)	(0.001)	(0.001)
Closed roof (base)				
No precipitation	−0.095*	−0.035	−0.076*	−0.014
	(0.056)	(0.063)	(0.040)	(0.047)

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TABLE 3 (Continued)

Dependent Variable: Concussion indicator	(1)	(2)	(3)	(4)
Light rain	−0.085	−0.014	−0.044	0.023
	(0.075)	(0.082)	(0.057)	(0.059)
Moderate rain	−0.116	−0.058	−0.116*	−0.058
	(0.071)	(0.077)	(0.061)	(0.065)
Heavy rain	−0.049	0.009	−0.174*	−0.124
	(0.144)	(0.128)	(0.097)	(0.091)
Light snow	−0.096	−0.093	−0.106	−0.047
	(0.244)	(0.256)	(0.131)	(0.128)
Moderate snow			−0.309**	−0.237*
			(0.138)	(0.136)
Game FE	No	Yes	No	Yes
Season FE	Yes	Yes	Yes	Yes
Team FE	Yes	Yes	Yes	Yes
Observations	1282	1282	1896	1896
Adjusted R^2	0.016	0.022	0.023	0.027

Note: $*p < .1$; $**p < .05$; $***p < .01$. Standard errors clustered by team in parentheses. Sample excludes all "neutral-site" games, and games played during final week of each season; sample for models (1) and (2) also excludes games that do not satisfy both Conditions 1 and 2.

First, the amount of travel required by the visiting team is correlated with concussion: doubling the distance the away team needs to travel increases the probability of a concussion by roughly 0.7 percentage points. For example, the expected incidence of a concussion for the Pittsburgh Steelers would be 0.7 percentage points higher if it were to play the Dallas Cowboys (932 miles of travel required) than the Tennessee Titans (471 miles). Among other controls considered, the use of an artificial playing surface, instead of grass, increases the risk of concussion by 6–7 percentage points. Despite the dangers of artificial turf being well-known—particularly for soft-tissue joint injuries—and despite complaints from NFL players and their union (Tretter, 2020), 44% of the games within our sample were played on artificial turf. The team strategy variables are largely insignificant with the exception of playing against a team that uses the "Air Coryell" offense strategy, which lowers the incidence of concussion by about 9 percentage points. Others (e.g., Teramoto et al., 2015) have found a significant impact of style of play characterized by a hastened pace of game play often requiring offensive players to catch the football at a time where their bodies are in positions vulnerable to unforeseen contact (i.e., "slant" routes).

Since NFL games are played in a combination of indoor and outdoor venues, we study how weather conditions impact injuries. Consistent with Haider et al. (2018), we find that the probability of a concussion decreases with higher temperatures and in worse weather conditions. Moderate snow has a large impact on the incidence of a concussion (decreases in concussion rates of 23–31 percentage points). Heavy rain and moderate rain also decrease the chance of concussion but this result is statistically significant only when game fixed effects are not included. These results must be interpreted with care because two competing effects may influence the impact of weather on injury: first, weather can directly affect the risk of injury (e.g., precipitation causes sleek playing surfaces). Second, and in contrast to the first, contest theory suggests that inclement weather adds noise in the determination of the outcome, thereby lowering the incentive to invest in risky strategies. According to Lazear and Rosen (1981), contestants invest less effort when luck plays a larger role in selecting the winner. We are unable to separate the direct playing-condition effect from the indirect behavioral response to the increase in luck.[17]

Overall, the impact on concussions is small for most control variables compared to the swing variable. Although weather conditions may have a large impact, this could be due to a decreased incentive effect. This suggests that incentives play a large role in explaining concussions, especially for high-swing games that take place at the end of the season, relative to other determinants of concussion that have received much attention in the literature. For example, a game in which a win guarantees a team a playoff berth (i.e., a swing value of one) increases the probability of a concussion by roughly four times as much as the impact of playing on an artificial surface.

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5.3 | Robustness

Table 4 considers multiple extensions from our baseline model. For the sake of exposition, the table uses the full sample and reports only the swing coefficients and key information about each specification. Supporting Information S1: Appendix 4.1 presents the results with the coefficient estimates of all control variables. It also offers detailed explanations about the construction of alternative swing variables.

Column 1 reports the impact of mathematical certainty to distinguish those teams that still have a chance to achieve the desired outcome from those that do not. In our application, the desired outcome is to earn a playoff berth. We say that a team has clinched a berth if the team is certain to qualify for the playoffs independent of the outcomes of all other games for the remainder of the season, that is, independently of its own and other teams' future performances (Hannah et al., 2019). A team is eliminated if it cannot earn a berth independent of the outcomes of all other games for the remainder of the season. A team is "in-the-hunt" if the team has neither clinched a berth nor is eliminated. A game is said to be mathematically certain in the first two cases as the outcome of the game cannot influence whether the team qualifies for the playoffs. A team has no incentive to take risk to win a mathematically certain game. Let H_t^i denote a dummy that is equal to one if game t is not mathematically certain for team i. Using our notations, we have

$$H_t^i = \begin{cases} 1 \; if \; Z_t^i > 0 \\ 0 \; if \; Z_t^i = 0 \end{cases}$$

Mathematical certainty is a coarse measure of the swing value (Lahvička, 2015). Only 6.0% of team-games are mathematically certain, most of which occur by the 15th game of a season. This is in contrast with the swing value, which discriminates games throughout the season (see Figure 4). The impact of H_t^i is small (a coefficient −0.04) and insignificant. This confirms the rejection of the risk-escalation hypothesis of Hannah et al. (2019) and demonstrates the importance of discriminating within games that are still mathematically uncertain, as done when one uses the swing to measure game stakes.

TABLE 4 Robustness.

Dependent variable: Concussion indicator	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Mathematical uncertainty	−0.037									
	(0.098)									
Swing, equal win probability		0.088								
		(0.138)								
Swing, perfect foresight			0.266**							
			(0.106)							
Swing, rational updating				0.237**	0.145	0.270**	0.226**	0.225**		
				(0.108)	(0.135)	(0.118)	(0.106)	(0.106)		
Swing, sequentially assigned									0.159*	
									(0.083)	
Own-team swing										0.116
										(0.073)
Other controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Game FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Season FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Team FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1896	1896	1896	1896	1464	1852	1896	1896	1896	1896
Adjusted R^2	0.025	0.025	0.027	0.027	0.028	0.025	0.027	0.027	0.026	0.026

Note: $^*p < .1$; $^{**}p < .05$; $^{***}p < .01$. Standard errors clustered by team in parentheses.

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Columns 2–4 of Table 4 compare three variants of the swing variable using different forecasting models of future win probabilities (see Supporting Information S1: Appendix 4.2). The coefficient estimate is positive for all three measures but it is not statistically significant when we assume equal win probabilities (column 3). This attests to the importance of team-specific variations in win probabilities that impact a team's swing value. A strong team that is very likely to qualify for the playoff should have low swing value. This is not the case when we assume that all teams have a 50% chance to win each future game. Ignoring team heterogeneity in the construction of the swing variable introduces a measurement error. Columns 3 and 4 assume that teams have different strengths but differ in how team strength is updated after each game. A team that has already won several games is more likely to win again in the future.[18] We first assume that teams have perfect foresight of the outcome of future games and that teams perfectly anticipate how other opponents' strengths will change over the season. The third measure is the one considered in Table 2 and assumes that teams updated their beliefs of strengths after each game. Teams observe past strengths and anticipate, in order to compute their chance to earn a playoff berth, that strengths remain constant for the remainder of the season. The coefficient estimates are similar in the two "heterogenous-strength" models (0.27 and 0.24) and both values are statistically significant. Taken together, these three models suggest that team heterogeneity is important.

Columns 5–9 pursue two approaches to investigate the robustness of the final payoff approach. First, we use the notion of mathematical certainty to add observations not selected under conditions 1 and 2 or to delete suspicious observations from the full sample. This demonstrates that even though mathematical certainty is not powerful on its own, it can help refine our approximation of swing, $\hat{Z}_t^i$. Second, we use the sequential feature of the NFL playoff seeding process to compute the swing values. We briefly discuss each robustness case, and where necessary, present detailed explanations in Supporting Information S1: Appendices 2 and 4.

Column 5 adds to the restricted sample the observations of exact swing according to mathematical certainty ($H_t^i = 0 \Rightarrow Z_{it} \equiv 0$). Doing so increases the sample of observations from 1282 to 1464. Column 6 starts from the full sample but eliminates the observations that are mathematically certain but are still coded as uncertain according to the swing value ($H_t^i = 0$ and $\hat{Z}_t^i > 0$). There are 44 such observations. Instead of eliminating these suspicious observations, column 7 uses the entire sample but assigns the value zero to these observations, using the new swing variable $\tilde{Z}_t^i = H_t^i Z_t^i$. A concern with this approach is that the expectations of Z_t^i and $\tilde{Z}_t^i$ are not equal anymore. To address this concern, column 8 reweights the non-mathematically certain observations to maintain the conditional expectation constant. Column 9 exploits the sequential feature of the playoff seeding process, to compute the exact swing value for a subset of observations for which it is computationally possible to do so (the playoff seeding process is described in Supporting Information S1: Appendix 4.2). The coefficient estimates in these five robustness tests vary between 0.15 and 0.27 and all coefficient estimates are statistically significant at conventional levels with one exception.

We have also considered alternative definitions of the swing variable. Column 10 reports the results with a variable that is equal to one's own swing. The coefficient estimate for $\hat{Z}_t^i$ is 0.12 and it is marginally significant (a p-value of .12). Other definitions were explored (results not reported here) such as the opponent's swing alone (a coefficient estimate of 0.20 and statistically significant at the 10% level) and both own and opponent swings (coefficient estimates of 0.01 and 0.20 for own and opponent swings with only the latter being statistically significant at the 10% level). Taken together, the evidence suggests both own and opponent swings increase concussions, with the latter having a larger impact on concussion than the former. This is consistent with the existence of externalities in risk-taking, although this is a tentative conclusion, due to the marginal statistical significance when we try to separate own and opponent effects.

The analysis measures game stakes as the probability to qualify for the playoffs. Clearly, qualification is not the only reason for why a game is important to a team. Winning more games also increases the chance to secure a better seed in the playoffs. We extend the analysis by distinguishing the first and second seeds of each conference, which gives its holder a first-round bye or, alternatively stated, an automatic bid to the second round of the playoffs. Figure 2 helps understand why bye seeds generate incentives not captured by the swing variable. According to that figure, earning more than 11 wins has very little impact on playoff qualification. The chance to obtain a bye seed, however, increases significantly with the 12th and 13th win (the bye seed payoff function looks similar to the curve plotted on Figure 2 but shifted to the right by about 2.5 points). We create a new swing variable for bye seeds. We now have two swing variables: a team can qualify for the playoffs for any seed or for a bye seed. The coefficient estimate on the bye seed swing variable measures the marginal benefit of earning a bye seed on top of qualifying for the playoff. The coefficient estimate on the bye swing variable is positive but not significant at 10% threshold. We conclude that teams take more risk to qualify to the playoff qualification than to win a first-round bye.

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6 | CONCLUSION

We show that the incidence of concussions sustained during NFL regular-season games depend on the benefit that the competing teams receive from winning a game in terms of the incremental chance from qualifying for the playoff. A team's chance to suffers at least one concussion increases by 10 percentage points in a game where a win increases one of the team's odds of playoff selection by 40 percentage points. This is the first evidence in support of the risk-escalation hypothesis stating that injuries increase with competition stakes. This establishes a starting point to assess the role of risk-taking incentives on athletes' injuries. For example, one may conjecture that our measured impact of game stakes on concussions applies to other injuries beyond concussions, and to other sports beyond professional American football.

A methodological contribution of our study is to construct an incentive-based measure, which we call the swing, of the importance of a game for a team and to propose a computationally friendly method to approximate it. The swing measure varies across teams and games of a season because of (a) heterogeneity in team quality; (b) luck (i.e., game outcomes are uncertain); and, most importantly, (c) the non-linear nature of playoff contention. The results demonstrate the importance of accounting for unobserved team heterogeneity (e.g., strength, strategy, etc.), while at the same time accounting for how team quality modulates how the swing variable responds to past outcomes.

Although this paper studies concussions, the methodology can be deployed to answer other questions. One could, of course, study the prevalence of other injuries but, more generally, one could construct swing variables in other sports having a similar two-stage competition structure as American football (e.g., Champions League qualification in European soccer), while investigating other behavior (e.g., by players, teams, fans, advertisers, etc.) that should respond to changes in game stakes.

The risk-escalation hypothesis also offers a strong motivation to seriously consider the health cost of incentives. Sports and health professionals have searched for causal factors that explain injuries and that have actionable implications. According to the contingency view, injuries are the consequence of controllable external conditions (e.g., weather, surface, game rules).[19] However, many doubt that the margin of improvement from the contingency view will be sufficient to appease the critics who, for example, oppose youth participation in dangerous sports, or question the long-term viability of sports that have high life-altering injury rates (Bull, 2021).

Our evidence in support of the risk-escalation hypothesis offers a new way to think about injury policy, based on the premise that it is possible to prevent some sport injuries by incentivizing behavior. In return, one can draw lessons to inform policy and practice, best summarized in the context of American football as following: Coaches, administrators, and players should be made aware that, although most concussions are unfortunate and random accidents, they are also sometimes caused by controllable determinants. Long travel time by visiting team and artificial playing surfaces tend to increase the likelihood of concussion, and, as our research shows, concussions happen more frequently in high stake games, and in particular those game that determine whether a team is likely to qualify for the playoffs. Teams and players should carefully assess the injury cost of exerting maximal effort in these games. Organizers, such as the National Football League, should take heed that compliance and enforcement of in-game concussion detection should depend on a game's stakes (e.g., have more concussion spotters in high-swing games) and should regulate athlete incentives to de-escalate risk taking (e.g., long return-to-play rules and caps on financial bonuses in high-swing competitions).

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in Western Economic Association International Data and Code Repository Project openICPSR at https://doi.org/10.3886/E191501V4, reference number 191501.

ENDNOTES

[1] Numerous former athletes have suffered tragic health conditions (early onset dementia, amyotrophic lateral sclerosis, or depression leading to suicide) that have been linked to sport related concussions.

[2] We use the term “berth” to indicate the binary outcome of whether a team qualifies for the post-season playoffs and “seeding” to indicate a team's relative ranking, or “seed,” among the qualifying teams.

[3] The risk-escalation hypothesis holds constant the cost of taking risk and the impact of risk on the probability to win a game.

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[4] Our proposed analysis tests this hypothesis using team stakes; one could also use variation in individual player incentives (e.g., future contract negotiations, likelihood to hit production target, etc.).

[5] Lawrence et al. (2016) and Gregory-Smith (2020) report the incidence of injury in the NFL by type (see discussion in Supporting Information S1: Appendix A1).

[6] Kerr et al. (2016) suggest that the most common reasons players do not self-disclose a concussion are not wanting to leave the game, not wanting to let their team down, lack of awareness of a concussion, and underestimating the severity of concussions. Calandrillo (2005) argues that physicians hired by the team face the same conflict of interest.

[7] The literature has studied multiple modulating factors influencing concussion, including weather, altitude, player fatigue, player position, days of rest, away game, and game outcome.

[8] These calculations are based on Section 37 of the collective bargaining agreement of the NFL and its player's union, which reports the additional pay that players get for making the post-season.

[9] Public Broadcasting Service, "Frontline: Concussion Watch," available at https://www.pbs.org/wgbh/frontline/interactive/concussion-watch/.

[10] Another measure of injury cost is the number of games missed after a concussion event. A limitation with that measure, however, is that few concussions in our sample remove the injured player beyond the game in which the concussion was sustained (i.e., most concussion events remove the player for the remainder of the current game but no other games thereafter). Measuring the severity of a concussion by the number of games missed may under-state the gravity of a concussion event.

[11] Using a larger sample of seasons (2002–2019 instead of 2012–2015), to estimate the conditional playoff qualification probabilities delivers more precise estimates, with no concern of bias since the playoff qualification and format remains the same in 2002–2019.

[12] This paper assumes that only playoff qualification matters, greatly simplifying the analysis. In practice, team performance also determines seeding positions in the playoffs.

[13] While $A(W)$ can be easily estimated, this is not the case for $A^i_{t-1}(W)$ even for low t, because these payoff functions are conditional on teams and past game-outcomes.

[14] When condition 1 holds, the absolute error, $\left|Z_{it} - \hat{Z}_{it}\right|$, is bounded by $\varepsilon_{i,t} = Max_{W\in Z}\Pr\left(\tilde{w}^i_{t.T} = W - w^i_{1,t-1}\right) - Min_{W\in Z}\Pr\left(\tilde{w}^i_{t.T} = W - w^i_{1,t-1}\right)$.

[15] Past injuries lower the swing value for teams that are unlikely qualify for the playoff and increase it for those that have a good chance to qualify for the playoff.

[16] There are other candidate explanations for the difference. The restricted sample systematically eliminates games late in the season, when average swing values are typically higher (see Figure 1). It could be that the relationship between swing and concussion is non-linear or that it changes at the end of the season.

[17] We also consider the team's win probability (using each team's Elo value) as a control for competitive balance. The coefficient estimate is small and insignificant and adding this control does not change the coefficient on the swing variable.

[18] Supporting Information S1: Appendix 4.2 uses the Elo approach to update a team's strength and to compute the win probabilities. See Boulier and Stekler (2003) for a discussion of the predictive power of a similar approach used to construct the New York Time power score NFL forecasts.

[19] For example, we find that distance traveled, playing surface and weather matter in our application. Policy implications would require limiting distance travel, using turf surfaces, and closely enforcing concussion rules during good weather.

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